Protein Sequence Identification and Analysis of Glutaminase (GLS) in Osteosarcoma

A Case Study on Glutamine Metabolism

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Contents:

1. Abstract
2. Introduction
3. Objectives
4. Materials & Methods
    - 4.1 Protein Identification (UniProt)
    - 4.2 Homology Search (BLASTp)
    - 4.3 Multiple Sequence Alignment (Clustal Omega)
    - 4.4 Domain and Motif Prediction (InterPro/Pfam)
    - 4.5 Protein Structure Prediction (AlphaFold)
    - 4.6 Variant Effect Prediction (Ensembl VEP)
    - 4.7 Proteotypic Peptide Selection (PeptideAtlas)
5. Results
    - 5.1 Protein Identification
    - 5.2 Homology Search
    - 5.3 Multiple Sequence Alignment
    - 5.4 Domain and Motif Prediction
    - 5.5 Protein Structure Prediction
    - 5.6 Variant Effect Prediction
    - 5.7 Proteotypic Peptide Selection
6. Discussion
7. Conclusion
8. References

1. Abstract:

Osteosarcoma is an aggressive bone cancer that often affects young people. Tumor cells reprogram their metabolism to support rapid growth, and glutamine is one key nutrient in this process. Glutaminase (GLS) converts glutamine into glutamate and plays an important role in cancer cell metabolism. This case study analyzes the human GLS protein using bioinformatics tools (UniProt, BLASTp, Clustal Omega, InterPro, AlphaFold, Ensembl VEP, and PeptideAtlas) to identify conserved regions, structural features, variant impacts, and proteotypic peptides. Results show GLS is highly conserved, two missense variants (His461Leu and Ser482Cys) were identified that may affect function, and three proteotypic peptides were selected for targeted proteomics. These findings highlight the importance of GLS in osteosarcoma metabolism and show how bioinformatics tools can be used to study disease-related proteins.

2. Introduction:

Osteosarcoma is the most common type of bone cancer. It mainly affects teenagers and young adults and often spreads to other parts of the body, such as the lungs. Even with surgery and chemotherapy, patients face challenges like recurrence and drug resistance. Because of this, researchers are exploring new ways to understand and treat osteosarcoma.

One important feature of cancer cells is metabolic reprogramming, where they change how nutrients are used compared to normal cells. In osteosarcoma, glutamine is a critical nutrient because it provides both energy and building blocks for rapid cell growth.

The enzyme Glutaminase (GLS) plays a central role in glutamine metabolism. GLS converts glutamine into glutamate, which enters the TCA cycle for energy production and supports the synthesis of molecules needed for cell survival. Due to this role, GLS has become an attractive target in cancer therapy, and inhibitors of GLS are already being tested in different cancers.

This case study focuses on the protein sequence identification and analysis of GLS in osteosarcoma. Using bioinformatics tools, the study aims to:

- identify important regions of the GLS protein,
- compare it with homologous proteins from other species,
- analyze its predicted 3D structure, and
- select proteotypic peptides useful for experimental detection.

3. Objectives:

The main aim of this case study is to understand how to identify and analyze a protein sequence using bioinformatics tools, focusing on Glutaminase (GLS) in osteosarcoma.

Specific objectives:

- Retrieve the GLS protein sequence from UniProt.
- Compare GLS with homologous proteins using BLASTp and sequence alignment.
- Identify important regions such as domains, motifs, and active sites.
- Visualize the 3D structure of GLS using AlphaFold.
- Predict the effects of GLS variants using Ensembl VEP.
- Select short proteotypic peptides for proteomics experiments using PeptideAtlas.
- Relate GLS sequence and structure to its role in osteosarcoma metabolism and therapy.

4. Materials & Methods:

This case study was carried out entirely using bioinformatics tools and databases. The analysis steps and methods are listed below.

4.1 Protein Sequence Retrieval:

- Database Used: UniProt (https://www.uniprot.org)
- Reason: UniProt is the most trusted and widely used protein database. It provides reviewed protein
- sequences (Swiss-Prot) along with detailed annotations like function, domains, isoforms, and post
- translational modifications.
- Protein: GLS (Glutaminase 1), Human
- Accession ID: O94925
- Sequence length: ~669 amino acids
- Information obtained: sequence (FASTA), isoforms, molecular weight, function

4.2 Homology Search (BLASTp):

- Tool: BLASTp (NCBI – https://blast.ncbi.nlm.nih.gov)
- Purpose: To find proteins similar to human GLS and check sequence conservation.
- Input: GLS sequence (FASTA, ID: O94925) from UniProt.
- Database: Non-redundant (nr) protein database.
- Parameters: Default (E-value threshold = 1e-5).
- Output: List of homologous proteins with % identity, species, alignment length, and E-value.

4.3 Multiple Sequence Alignment (MSA):

- Tool: Clustal Omega (https://www.ebi.ac.uk/Tools/msa/clustalo/)
- Purpose: To align GLS sequences and find conserved regions across isoforms/species.
- Input: FASTA sequences of human GLS and homologs (from BLAST results).
- Parameters: Default settings.
- Output: Alignment showing conserved residues (marked with "*") and similarities between isoforms.

4.4 Domain and Motif Analysis:

- Tools: InterPro (https://www.ebi.ac.uk/interpro/) and Pfam (https://pfam.xfam.org)
- Purpose: To find functional regions of GLS, including domains and motifs.
- Input: GLS FASTA sequence (ID: O94925).
- Process: Sequence submitted to InterPro/Pfam using default settings.
- Output: Identified Glutaminase domain and conserved motifs with a diagram showing domain locations.

4.5 3D Structure Analysis:

- Database: AlphaFold Protein Structure Database (https://www.alphafold.ebi.ac.uk)
- Purpose: To study the 3D structure of GLS and see domain arrangement.

- Input: UniProt ID O94925 (GLS).
- Process: Search in AlphaFold and view/download predicted model.
- Output: 3D structure with pLDDT confidence scores, showing catalytic pocket and conserved regions.

4.6 Variant Effect Prediction:

- Tool: Ensembl Variant Effect Predictor (VEP) (https://www.ensembl.org/vep)
- Purpose: To predict the effect of GLS variants on protein function.
- Variants studied: p.His461Leu and p.Ser482Cys.
- Process: Entered into VEP using protein-level notation with default transcript set.
- Output: Classification of variants (e.g., missense) and predicted functional impact

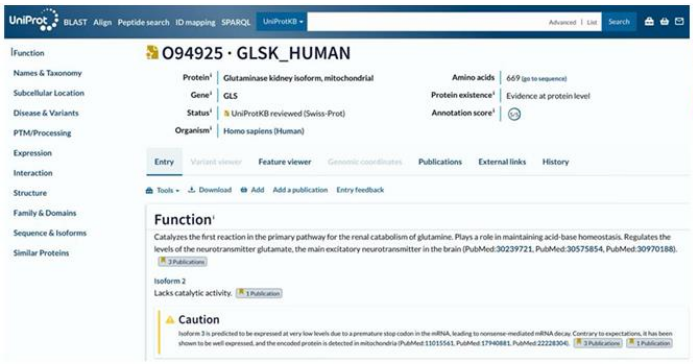
4.7 Proteotypic Peptide Selection

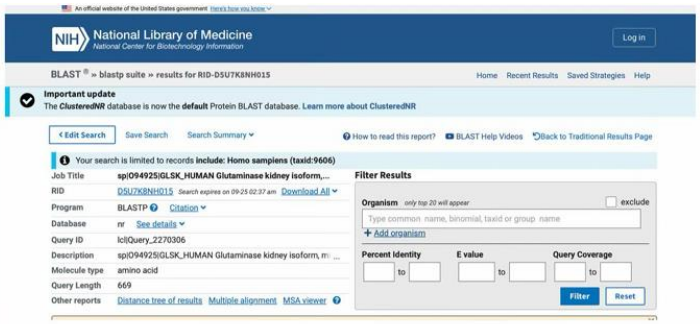
- Database: PeptideAtlas (http://www.peptideatlas.org)
- Purpose: To identify short, unique peptides of GLS useful for detection in proteomics.
- Input: GLS protein sequence (FASTA, ID: O94925).
- Process: Sequence searched in PeptideAtlas; peptides predicted for MS detection.
- Output: Unique GLS peptides (7–25 aa) suitable for proteomics detection.

5. Results:

5.1 Protein Sequence Retrieval (UniProt):

The protein sequence of human Glutaminase (GLS) was retrieved from the UniProt database. The reviewed entry with accession ID O94925 was selected. The entry shows that GLS is a mitochondrial enzyme with a sequence length of about 669 amino acids. The canonical sequence in FASTA format was downloaded for further analysis.





5.2 Homology Search (BLASTp):

A BLASTp search was performed using the human Glutaminase (GLS) protein sequence. The results showed that the top matches were all human isoforms of GLS, including mitochondrial isoforms 1, 2, 3, and 4. All isoforms had very high similarity (≥99% identity) with an E-value of 0.0, confirming that they are closely related variants of the same protein. This shows that GLS exists in multiple isoforms in humans, which may have slightly different roles in different tissues, but they all share the same core glutaminase function.

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
glutaminase kidney isoform, mitochondrial isoform 1 precursor [Homo sapiens]	Homo sapiens	1393	1393	100%	0.0	100.00%	669	NP_055720.3
glutaminase [Homo sapiens]	Homo sapiens	1391	1391	100%	0.0	99.85%	669	AAG47842.1
unnamed protein product [Homo sapiens]	Homo sapiens	1391	1391	100%	0.0	99.85%	669	BAF82436.1
glutaminase, isoform CRA_e [Homo sapiens]	Homo sapiens	1344	1344	100%	0.0	97.31%	651	EAX10860.1
glutaminase kidney isoform, mitochondrial isoform 3 [Homo sapiens]	Homo sapiens	1179	1179	94%	0.0	94.97%	661	NP_001424211.1
glutaminase kidney isoform, mitochondrial isoform 2 [Homo sapiens]	Homo sapiens	1148	1148	82%	0.0	100.00%	598	NP_001243239.1
Chain A, Glutaminase kidney isoform, mitochondrial [Homo sapiens]	Homo sapiens	1146	1146	82%	0.0	99.82%	598	8SZJ_A
glutaminase C [Homo sapiens]	Homo sapiens	1144	1144	82%	0.0	99.64%	598	AAD47056.1
glutaminase kidney isoform, mitochondrial isoform 4 [Homo sapiens]	Homo sapiens	1144	1144	82%	0.0	99.82%	559	NP_001424212.1

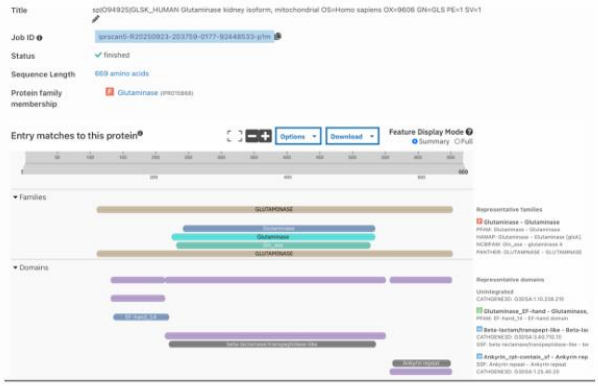
5.3 Multiple Sequence Alignment (MSA):

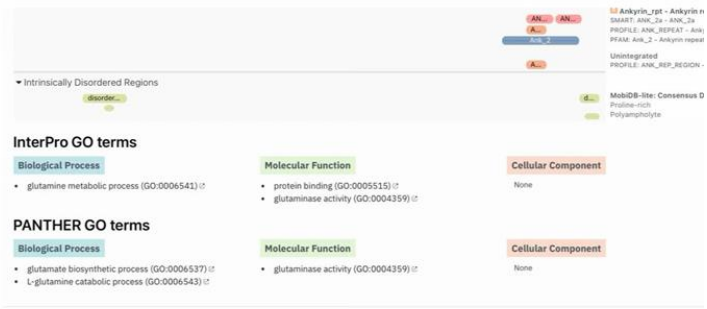
The amino acid sequences of three human GLS isoforms were aligned using Clustal Omega. The alignment output showed that most residues were highly conserved, which was indicated by the "*" symbols across the sequences. This means that the functional and catalytic regions of GLS remain unchanged between isoforms, even though small variations are present in certain regions. These conserved parts are important because they help maintain the essential activity of the enzyme.

```
BAF82436.1      MLRLTLQTTSDGVMLDKDLFKKCVQSNIVLLTQAFRRKFVIPDFMSFTSHIDELYESAKK   240
NP_055720.3     MLRLTLQTTSDGVMLDKDLFKKCVQSNIVLLTQAFRRKFVIPDFMSFTSHIDELYESAKK   240
AAG47842.1      MLRLTLQTTSDGVMLDKDLFKKCVQSNIVLLTQAFRRKLVIPDFMSFTSHIDELYESAKK   240
                ************************************:*******************

BAF82436.1      QSGGKVADYIPQLAKFSPDLWGVSVCTVDGQRHSTGDTKVPFCLQSCVKPLKYAIAVNDL   300
NP_055720.3     QSGGKVADYIPQLAKFSPDLWGVSVCTVDGQRHSTGDTKVPFCLQSCVKPLKYAIAVNDL   300
AAG47842.1      QSGGKVADYIPQLAKFSPDLWGVSVCTVDGQRHSTGDTKVPFCLQSCVKPLKYAIAVNDL   300
                ************************************************************

KSGVAGGILLVVPNVMGMMCWSPPLDKMGNSFKGIHFCHDLVSLCNFHNYDNLRHF
KSGVAGGILLVVPNVMGMMCWSPPLDKMGNSVKGIHFCHDLVSLCNFHNYDNLRHF
KSGVAGGILLVVPNVMGMMCWSPPLDKMGNSVKGIHFCHDLVSLCNFHNYDNLRHF
********************************.************************
```

5.4 Domain &Motif Analysis:

Domain analysis of GLS using InterProScan revealed that the protein belongs to the Glutaminase family (IPR015868). The sequence contains a major Glutaminase catalytic domain, which is responsible for converting glutamine into glutamate. Additional motifs such as EF-hand (calcium-binding) and ankyrin repeats (protein interactions) were also detected. The functional annotations confirm that GLS is involved in glutamine metabolism and glutamate biosynthesis, which are essential for cancer cell growth and survival.





5.5 3D Structure (AlphaFold):

- The predicted 3D structure of human Glutaminase (GLS) was retrieved from the AlphaFold Protein Structure Database (UniProt ID: O94925).
- The model consists of 669 amino acids.
- The average pLDDT confidence score was 80.23 (High).
- Most of the regions were predicted with very high accuracy(blue),while only a few flexible regions showed lower confidence(orange/red).
- The protein is organized into three domains (TED01, TED02, TED03):
  - Domain 1: residues 138–215 (pLDDT ~91.3)
  - Domain 2: residues 230–523 (pLDDT ~96.3, highest confidence)
  - Domain 3: residues 545–646 (pLDDT ~90.6)
- These results confirm that the core catalytic glutaminase domain is highly stable and conserved, supporting its essential role in glutamine metabolism and cancer related pathways.

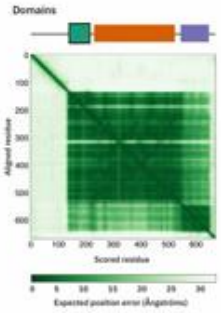
Glutaminase kidney isoform, mitochondria...
AF-O94925-F1-v4 • Google DeepMind dataset
Tell us what you think of the new look
Share your feedback
Summary and Model Confidence
Domains
Annotations
Similar Proteins
Protein
Glutaminase kidney isoform, mitochondrial
Gene
GLS
Source organism
Homo sapiens
UniProt
O94925
Biological function
Data unavailable
Experimental structures
None available in the PDB
Average pLDDT
80.23 (High)
pLDDT distribution
67% Very high
7.9% High
2.1% Low
23% Very low

X Axis

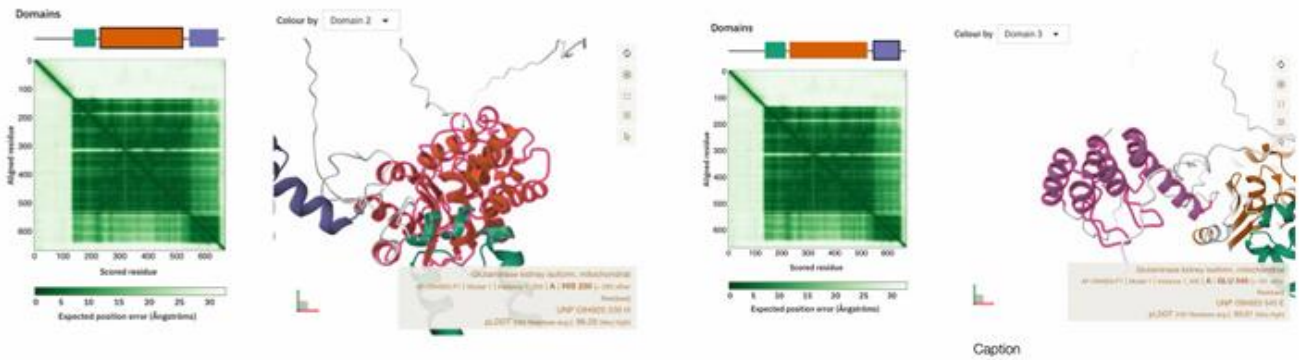
Y Axis

Z Axis

TED Consensus Domains
TED Results


Domains
Aligned residue
Scored residue
Expected position error (Ångström)

Colour by
Domain 1



5.6 Variant Effect Prediction:

In this step, the Ensembl Variant Effect Predictor (VEP) was applied to study possible effects of changes in the GLS (Glutaminase) protein sequence. Two variants were analysed:

- GLS:p.His461Leu
- GLS:p.Ser482Cys

The VEP results showed that both variants are missense mutations, which means the amino acid sequence of the protein is changed:

- His461Leu → codon change from CAT → CTT (Histidine to Leucine)
- Ser482Cys → codon change from TCT → TGT (Serine to Cysteine)

These variants are also recorded in public databases (dbSNP IDs: rs1558986214 and rs1558986215). The pie chart and results table from VEP confirm that 100% of these variants are missense. Missense mutations can affect protein structure and function, which is relevant in understanding their possible role in bone cancer.

Show/hide columns (14 hidden)

Uploaded variant	Location	Allele	Consequence	Symbol	Gene	Feature type	Feature	Biotype	Exon	cDNA position	CDS position	Protein position	Amino acids	Codons
GLS:p.His461Leu	2:190927439-190927439	T	missense_variant	GLS	ENSG00000115419	Transcript	ENST00000320717.8	protein_coding	12/18	1646	1382	461	H/L	CAT/CTT
GLS:p.His461Leu	2:190927439-190927439	T	missense_variant	GLS	ENSG00000115419	Transcript	ENST00000338435.9	protein_coding	12/15	1646	1382	461	H/L	CAT/CTT
GLS:p.His461Leu	2:190927439-190927439	T	missense_variant	GLS	ENSG00000115419	Transcript	ENST00000950145.1	protein_coding	12/19	1646	1382	461	H/L	CAT/CTT
GLS:p.His461Leu	2:190927439-190927439	T	missense_variant	GLS	ENSG00000115419	Transcript	ENST00000950146.1	protein_coding	12/16	1645	1382	461	H/L	CAT/CTT
GLS:p.Ser482Cys	2:190930456-190930456	G	missense_variant	GLS	ENSG00000115419	Transcript	ENST00000320717.8	protein_coding	13/18	1709	1445	482	S/C	TCT/TGT
GLS:p.Ser482Cys	2:190930456-190930456	G	missense_variant	GLS	ENSG00000115419	Transcript	ENST00000338435.9	protein_coding	13/15	1709	1445	482	S/C	TCT/TGT

5.7 Proteotypic Peptide Selection:

PeptideAtlas was used to check the GLS protein (ID: O94925). Some short peptide sequences were found that are best for experiments because they are unique and easy to detect. The main selected peptides are:

- LTLQTTSDGVMLDK
- MAGNEYVGFSNATFQSER
- GSTHPQPGVSPPAAPGPK

These peptides are reliable and can be used for protein detection in lab studies.

show column descriptions

Show more rows]

Accession	Pre AA	Sequence	Fol AA	PSS	PSieve	ESPP	DPred	APEX	STEPP
PAp00074233	R	LTLQTTSDGVMLDK	D	0.95	0.96	0.73	0.70	0.25	0.77
PAp00074509	K	MAGNEYVGFSNATFQSER	E	0.95	0.98	0.22	0.73	0.28	0.39
PAp00034352	K	GSTHPQPGVSPPAAPAAPGPK	D	0.95	0.95	0.52	0.54	0.24	0.76
PAp00133441	R	TALHVAAAEGHVEVVK	F	0.93	0.95	0.45	0.78	0.08	0.73
	R	GLSSSPSEILQELGK	G	0.93	0.97	0.78	0.73	0.05	0.74
PAp00072839	K	LFLNEDDKPHNPMVNAGAIVVTSLIK	Q	0.92	0.93	0.05	0.75	0.15	0.17
	K	ILQEYQVQYTPQGDSDNGK	E	0.91	0.96	0.41	0.71	0.02	0.76
PAp00130937	K	QGLLPSLEDLLFYTIAEGQEK	I	0.90	0.99	0.04	0.64	0.00	1.02
PAp00081454	K	YAIAVNDLGTEYVHR	Y	0.87	0.91	0.25	0.63	0.03	0.62
PAp00010129	K	FSPDLWGVSVCTVDGQR	H	0.87	0.21	0.25	0.82	0.48	-0.35
	R	LHPWWGGGGWPAEPLAR	G	0.87	0.95	0.29	0.54	0.06	0.11
PAp00064881	K	DGPGETDAFGNSEGK	E	0.78	0.70	0.26	0.76	0.00	0.79
PAp00131924	K	FVIPDFMSFTSHIDELYESAK	K	0.68	0.68	0.03	0.69	0.00	0.59
PAp00566165	R	VLSPEAVR	N	0.60	0.02	0.80	0.64	0.33	0.34
PAp00131789	K	FDYVMQFLNK	M	0.57	0.09	0.11	0.62	0.33	0.29
PAp00353322	R	FALSAMDMEQR	D	0.53	0.54	0.63	0.62	0.02	0.19
PAp00384029	K	VADYIPQLAK	F	0.49	0.26	0.82	0.56	0.14	0.62
	R	GGGRPAAGPAAAAR	L	0.41	0.51	0.32	0.71	0.00	-0.07
PAp00133636	R	WNNTPMDEALHFGHHDVFK	I	0.31	0.43	0.13	0.49	0.03	0.41
PAp00031550	K	SVINLLFAAYTGDVSALR	R	0.27	0.36	0.09	0.73	0.00	0.33

Table: Selected Proteotypic Peptides of GLS

Peptide Sequence	Length (aa)	Notes
LTLQTTSDGVMLDK	14	Unique, detectable fragment
MAGNEYVGFSNATFQSER	18	High confidence peptide
GSTHPQPGVSPPAAPGPK	18	Reliable marker for GLS

Discussion:

The study focused on the GLS (Glutaminase) protein, which plays an important role in the glutamine metabolism pathway. Cancer cells, including osteosarcoma cells, use large amounts of glutamine as fuel for growth and survival. GLS is the enzyme that converts glutamine into glutamate, which then enters different metabolic pathways that support tumor energy, cell division, and resistance to stress.

From the bioinformatics analysis, several important points were observed. The BLASTp and multiple sequence alignment results showed that GLS is highly conserved across species, which means its function is very important for life. InterPro results confirmed the presence of catalytic domains needed for enzyme activity, while AlphaFold provided the 3D structure that helps in visualizing how these domains fold and interact. These features explain why GLS is essential for metabolism.

Variant analysis using VEP identified two missense mutations (His461Leu and Ser482Cys). Such changes can alter protein structure and may make GLS more active or unstable. In cancer, gain-of-function changes in GLS can increase glutamine breakdown, giving cancer cells more resources to grow. This supports the idea that GLS mutations or overactivity could be linked to osteosarcoma progression, since this cancer type depends on altered metabolism.

Finally, proteotypic peptide selection from PeptideAtlas identified short GLS peptide sequences that can be detected in mass spectrometry. These peptides are useful for future laboratory experiments where GLS needs to be measured in cancer samples.

Overall, the analysis highlights that GLS is not only a normal metabolic enzyme but also a potential cancer-related protein. Its role in glutamine metabolism makes it relevant to osteosarcoma, where tumor cells depend on glutamine for rapid growth. Understanding GLS variants, structure, and peptides can help in future research, diagnostics, and targeted therapies.

Conclusion:

The GLS protein is a key enzyme in glutamine metabolism and is linked to cancer cell growth, including osteosarcoma. This case study demonstrated the use of bioinformatics tools to explore its sequence, structure, variants, and proteotypic peptides. The results indicate that GLS variants may influence cancer progression, while the identified peptides provide useful markers for future detection and measurement of GLS. Overall, the study highlights GLS as an important protein in osteosarcoma research and a potential target for therapeutic strategies.

References

A) Research Articles

- Ren L, Ruiz-Ruiz C, Li X, et al. Glutaminase-1 (GLS1) inhibition limits metastatic progression in osteosarcoma.Cancer & Metabolism. 2020; 8(1):4. doi:10.1186/s40170-020-0209-8
- Shi Y. Metabolic reprogramming in osteosarcoma. Frontiers in Oncology. 2023; 13:12118251. doi:10.3389/fonc.2023.12118251

B) Databases and Tools

- UniProt Database – https://www.uniprot.org
- NCBI BLASTp – https://blast.ncbi.nlm.nih.gov
- Clustal Omega (EBI) – https://www.ebi.ac.uk/Tools/msa/clustalo/
- InterPro Database – https://www.ebi.ac.uk/interpro/
- AlphaFold Protein Structure Database – https://www.alphafold.ebi.ac.uk
- Ensembl Variant Effect Predictor (VEP) – https://www.ensembl.org/vep
- PeptideAtlas – http://www.peptideatlas.org